Zhaoqi Song

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EDUCATION

Xiamen University *Expected June 2027*
MEng of Bioengineering
Advisor: *Prof.* Chaoyong Yang

Fujian Medical University *June 2024*
Bachelor of Bioinformatics
GPA: 3.51 / 4.0

BACKGROUND

Interests: Foundation Model, Virtual Cell, AI for Biomedicine and Bioinformatics
Skills: Python, R, Linux Shell, PyTorch; *Claude Code / Codex*
Language: English **(IELTS - 7.0, TOEFL - 5.0)**

INTERNSHIP

The Hong Kong Polytechnic University *2026.9 – Present*
Research Intern | Hong Kong S.A.R.
Collaborating with Dr.Du Ye and Qian Niu.

The University of Tokyo *2026.9 – Present*
Research Intern | Tokyo, Japan
Working with Qian Niu at Matsuo-Iwasawa Lab on Foundation Model-based Virtual Cell Construction.

PUBLICATIONS

Peer-Reviewed Publications

[1] Wang M*, **Song Z***. (2026). DeepALM: A Context-Aware Deep Learning Framework for Antimicrobial Peptide Prediction. *Current Bioinformatics*. http://www.eurekaselect.com/article/156631

[2] Wang M*, **Song Z***. (2024). Depression-associated gut microbes, metabolites and clinical trials. *Frontiers in Microbiology*, https://doi.org/10.3389/fmicb.2024.1292004

[3] Sun H*, **Song Z***. (2023). MMiKG: a knowledge graph-based platform for path mining of microbiota–mental diseases interactions. *Briefings in Bioinformatics*, https://doi.org/10.1093/bib/bbad340

[4] Liang J, Huang F, **Song Z**, et al. (2023). Impact of NAD+ metabolism on ovarian aging. *Immunity & Ageing*, https://doi.org/10.1186/s12979-023-00398-w

Preprints

[5] **Song Z***, et al. (2026). scAnyOmics: Foundation-Model Distillation for Any-to-Any Completion of Multi-omics Virtual Cells, *arXiv*

[6] **Song Z***. ProCLIP: Contrastive Multimodal Prediction of Spatial Proteomics from H&E and Spatial Transcriptomics.

[7] Zou Y*, **Song Z***. SMART: Super-resolution Multimodal Transcriptomics for spatial RNA sequencing.

Patents

[8] **Song Z**, Sun H, Chen Q, Wang M, Yang F. (2023). A knowledge graph construction method targeting the gut-brain axis. China Patent No. 202310235160.3., https://www.docin.com/p-4551655414.html

(indicates co-first author)*

PROFESSIONAL EXPERIENCE

The University of Tokyo, Matsuo-Iwasawa Lab **Tokyo, Japan**
Research Project *2026.9 - Present*
Project Theme: "scAnyOmics: A Foundation Model for Multimodal Virtual Cells and Perturbation Prediction"
Supervisor: *Prof.* Qian Niu

- Developing knowledge-distillation Foundation model to integrate scRNA-seq, ADT, and ATAC
- Missing-modality completion, perturbation effects, and continuous-dose responses
- Designing validation across healthy cohorts and out-of-distribution disease cohorts

Xiamen University, Tan Kah Kee Innovation Laboratory **Xiamen, China**
Research Project *2026.1 - Present*
Project Theme: "ProCLIP: Contrastive Multimodal Prediction of Spatial Proteomics from H&E and Spatial Transcriptomics"
Supervisor: *Prof.* Chaoyong Yang

- Developed a multimodal framework integrating ST, H&E, and imaging mass cytometry (IMC)
- Virtual Proteomics construction to predict missing protein channels
- Facilitated downstream biomedical analyses in spatial proteomic heterogeneity and cell–cell interactions

Xiamen University, Tan Kah Kee Innovation Laboratory **Xiamen, China**
Research Project *2025.7 - Present*
Project Theme: "*SMART*: Multimodal–Based Super-Resolution Enhancement for Spatial Omics"
Supervisor: *Prof.* Chaoyong Yang

- Developed a multimodal framework for high-resolution gene expression mapping using in situ Sequence (ISS)
- Enabled accurate cell-type annotation and tissue structure delineation across heterogeneous tumor samples
- Supported biomedical discovery by revealing spatial heterogeneity and intercellular communication

AWARDS

- ✓ **Wang Deyao Outstanding Student Scholarship**, Xiamen University
- ✓ **Outstanding Graduate**, Fujian Medical University
- ✓ **Outstanding Student Scholarship**, Fujian Medical University
  – *Second-Class Scholarship*, 2023
  – *Third-Class Scholarship*, 2024
- ✓ **Merit Student**, Fujian Medical University
- ✓ **Mathematical Contest in Modeling (MCM/ICM)**, Honorable Mention (Top 20%), 2023
- ✓ **National Undergraduate Mathematical Contest in Modeling (CUMCM, Higher Education Press)**
  – *Provincial First Prize*, 2023
- ✓ **FLTRP • Cup National English Public Speaking Contest** – *Provincial First Prize*, 2022

WORKSHOP AND CONFERENCE

- ✓ **Peking University Life Science Interdisciplinary Summer Camp**, 2026, Beijing, China
- ✓ **Chinese Institute for Brain Research (CIBR) Winter School**, 2025
  - *Peking University, Beijing, China*
- ✓ **Visiting Research Intern**, Department of Bioinformatics, *Prof. Yong Zhang* Laboratory, 2023
  - *Tongji University, Shanghai, China*
- ✓ **Outstanding Undergraduate Summer Camp**, Shanghai Institute of Precision Medicine (SHIPM), 2023
  - *Shanghai Jiao Tong University, Shanghai, China*
- ✓ **Outstanding Undergraduate Summer Camp**, Liangzhu Laboratory, 2023
  - *Zhejiang University, Hangzhou, China*
- ✓ **Outstanding Undergraduate Summer Camp**, School of Basic Medical Sciences, 2023
  - *Southern Medical University, Guangzhou, China*